

**INCLUSIVE MOBILE LEARNING APP FOR VISUALLY,
HEARING, PHYSICALLY AND COGNITIVELY
IMPAIRED STUDENTS (GRADE 3-5)**

25-26J-207

Project Proposal Report

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BSc (Hons) in Information Technology Specializing in Information
Technology

Department of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Date:

Signature of the supervisor:

Date:

ABSTRACT

Physical disabilities present significant barriers for primary school students in grades 3-5, limiting their ability to interact with traditional touch-based educational applications and hindering independent learning. Existing assistive technologies often focus on single input methods and lack comprehensive solutions tailored for young learners with diverse physical impairments. To address this gap, this research proposes an inclusive mobile learning application developed with Flutter that integrates multiple adaptive interaction methods including voice-controlled navigation, dwell-time activation, gaze tracking, and facial gesture recognition. The system incorporates customizable accessibility features such as adjustable button sizes, dwell time duration, and visual feedback mechanisms to accommodate individual motor abilities and preferences. Safety features including emergency voice commands, visual countdown indicators, and accidental touch cancellation ensure secure and frustration-free interaction for young users. Additionally, the application provides an AI-powered assistant for learning support, achievement-based motivation systems with collectibles, and real-time progress tracking to enhance engagement and educational outcomes. Cross-platform compatibility ensures accessibility across Android and iOS devices, while integration with assistive hardware through Bluetooth connectivity expands interaction possibilities. By combining multi-modal input methods, personalized customization options, and child-friendly design principles, the proposed solution delivers a comprehensive and empowering educational platform that enables students with physical disabilities to access digital learning content independently, safely, and effectively.

Keywords: Physical disabilities, Motor impairments, Inclusive education, Mobile learning application, Flutter, Multi-modal interaction, Voice control, Dwell-time activation, Gaze tracking, Facial gesture recognition, Assistive technology, Accessibility, Cross-platform development, AI assistance, educational technology, independent learning, Safety features, Customizable interface

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AT	Assistive Technology
API	Application Programming Interface
AI	Artificial Intelligence
iOS	iPhone Operating System
WCAG	Web Content Accessibility Guidelines
AWS	Amazon Web Services
UI/UX	User Interface/User Experience
BSc	Bachelor of Science

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1. INTRODUCTION

In today's digital learning environment, accessibility is not just a feature, it is a necessity. Students with physical disabilities often face significant challenges in navigating and interacting with traditional mobile applications, which can limit their ability to learn effectively and independently. My part of the project focuses on designing and implementing an inclusive interface within our e-learning mobile app, specifically tailored for students with physical impairments in grades 3-5.

Students with physical disabilities in this age group represent a particularly vulnerable population in the digital divide. At this critical developmental stage, children are building foundational academic skills while developing their sense of independence. Existing assistive technologies, while valuable, often suffer from critical limitations that prevent them from serving as comprehensive solutions. Most current solutions focus on addressing single interaction modalities such as switch access or voice commands without considering the complex needs of students who may benefit from multiple alternative input methods. Furthermore, many assistive technologies are designed for adult users, lacking the child-friendly interfaces and age-appropriate content that younger learners require.

This section of the application will integrate multiple adaptive interaction methods to ensure that students with limited hand or finger mobility can still engage with lessons, complete activities, and navigate the platform with ease. Core functionalities such as one-switch and voice-controlled navigation, large touch-friendly buttons, and Bluetooth assistive device compatibility will be combined with advanced features like dwell-time activation, gaze tracking, and facial gesture controls. These features will allow students to interact using various body parts or alternative input methods, reducing dependency on fine motor skills and providing multiple pathways for successful interaction.

Additionally, the interface will support customizable settings such as dwell time duration, button size, and feedback type to adapt to the unique needs of each learner. Students with physical disabilities represent a highly diverse group with varying degrees of motor control and preferred interaction methods, making personalization essential for effective learning. Safety and error prevention are also prioritized through visual countdown indicators, the ability to cancel accidental touches, and an emergency voice or sound-based stop command to ensure secure and frustration-free interaction for young users.

The technical implementation leverages cutting-edge technologies including Flutter for cross-platform development, ensuring compatibility across Android and iOS devices. Advanced AI models from Hugging Face enhance speech recognition accuracy for children's voices, improve gaze estimation precision, and enable sophisticated facial

gesture recognition. Real-time data processing provides immediate feedback and accurate grading for student activities, while achievement-based motivation systems with collectible rewards encourage continued engagement with subject-related activities.

By blending practicality with innovative control methods, this part of the project aims to create a truly accessible, comfortable, and empowering learning experience for students with physical disabilities. The integration of a friendly AI assistant provides contextual help and support, reducing dependence on external assistance and promoting independent learning. Through this comprehensive approach, the application will demonstrate how multiple assistive technologies can be seamlessly integrated into mainstream educational platforms, ultimately creating a learning environment where physical disabilities do not determine educational outcomes.

1.1 BACKGROUND

Inclusive education is recognized as a fundamental right, yet students with physical disabilities in Sri Lanka and globally continue to face significant barriers in accessing digital learning tools. According to national disability statistics, more than 390,000 students in Sri Lanka live with disabilities, many of whom are of school-going age. Existing educational applications are often developed for a single disability type such as visual or hearing impairments and rarely address the specific challenges faced by physically impaired learners, including limited motor control or reliance on assistive devices. In addition, most applications are not optimized for younger learners in Grades 3–5, often lack child-friendly interfaces, and provide little to no support for local languages such as Sinhala. These limitations reduce accessibility and restrict meaningful participation in digital learning environments.

To bridge this gap, the proposed research aims to design and develop a mobile application module specifically tailored for students with physical disabilities in Grades 3–5. The solution integrates multiple accessibility features into a single, child-friendly platform, including voice commands, large interactive buttons, and Bluetooth-based assistive device compatibility. By localizing the interface and aligning it with the Sri Lankan primary school curriculum, the system seeks to enhance accessibility, engagement, and inclusion. Through this approach, the research aspires to contribute to a more equitable learning environment where all students can actively participate regardless of their physical limitations.

1.2 LITERATURE SURVEY

- i. **Borgestig, Rytterström & Hemmingsson (2016) – Eye Gaze Performance**
The longitudinal analysis confirms the feasibility of gaze-based AT for children with severe impairments. However, it focuses solely on eye gaze. Unlike this project, it does not integrate multiple adaptive controls, curriculum alignment, or customizable dwell-time and child-specific safeguards.
- ii. **Borgestig et al. (2017) – Gaze-Based AT in Daily Activities (Intervention_Study)**
Shows positive impact of gaze-based technology in everyday tasks but limits scope to functional independence. The proposed project extends this by targeting educational outcomes with multimodal input, motivational rewards, and real-time learning support.
- iii. **Borgestig, Rytterström & Hemmingsson (2017) – Parents’ Experiences**
Highlights parental observations of gaze AT in daily use, noting usability and empowerment benefits. Yet it lacks structured educational frameworks, safety/error prevention, and multimodal pathways. The project addresses these omissions by embedding child-centered learning workflows and protective mechanisms.
- iv. **Hsieh (2024) – Feasibility of Eye-Gaze Intervention**
Assesses feasibility of eye-gaze interventions in schools, showing promise for students with severe motor and communication difficulties. However, it remains limited to gaze. The project advances feasibility into a full educational system by layering multiple alternative inputs and customizable features.
- v. **Wilson et al. (2022) – Eye-Gaze or Switch for Cognitive Testing**
Validates acceptability of gaze/switch-based controls for assessments in cerebral palsy. While important, it remains assessment focused. The proposed system expands to continuous educational engagement with multiple input methods, feedback, and progression tools.
- vi. **Andreassen et al. (2024) – Psychosocial Impact of Eye-Gaze Technology**
Finds positive psychosocial outcomes for children and adults using gaze AT. Unlike this project, it does not connect psychosocial benefits with formalized, safe, child-specific learning modules. The current solution unifies psychosocial empowerment with academic performance and adaptive navigation.

- vii. Hemmingsson et al. (2018) – Case Report: Eye-Gaze Control in Spinal_Injury**
Demonstrates early intervention benefits with gaze control in a single child. The scope is narrow and lacks system-level generalization. The proposed app generalizes these benefits into a scalable, child-friendly educational platform with multimodal and localized support.
- viii. Masayko et al. (2024) – Parents’ Perceptions of Eye-Gaze AAC**
Explores family perceptions of AAC through gaze technology. While emphasizing usability, it does not incorporate multiple modalities or structured pedagogy. This project embeds parental/teacher tracking within a multimodal, curriculum-aligned framework.
- ix. Hsieh et al. (2021) – Communicative Interaction with and Without Eye-Gaze**
Focuses on how gaze technology affects partner communication among children with complex needs. While informative for interaction, it does not address structured learning, gamification, or multimodal accessibility. The proposed app bridges this by applying gaze in formal curriculum-aligned education.
- x. Kairy et al. (2021) – Mobile App for Social Participation**
Presents a mobile app for physical disability support, validated for usability, but its focus is on social participation. Unlike this project, it lacks multimodal navigation, child-friendly learning design, and localized content.
- xi. Mott et al. (2018) – Smartphone Photography Accessibility**
Analyzes accessibility barriers for motor-impaired users in photography. Insights highlight UI limitations but do not target education or multimodal pathways. The proposed solution applies similar accessibility learnings but extends them into multimodal educational interaction.
- xii. Cicek et al. (2020) – Head-Based Pointing on Smartphones**
Develops and evaluates head-based pointing as an input method. While relevant to alternative access, it does not integrate multiple input streams, safety layers, or educational content. This project expands head/face-based gestures into a broader multimodal, child-centered learning system.

- xiii. **Fan et al. (2020) – Eyelid Gestures on Mobile Devices**
Proposes eyelid gesture interactions for motor-impaired users. Innovative but narrow in scope, with no child-oriented safety features or educational context. The project incorporates facial gestures among other modalities, adapted for children’s learning tasks.
- xiv. **Castellucci & MacKenzie (2019) – TiltWriter**
Explores tilt-based text entry without touch. While valuable, it is restricted to text input. The proposed solution adapts tilt into a multimodal educational framework with gamification and adaptive UI.
- xv. **Ljubic et al. (2014) – Predicting Tilt-Based Input**
Presents models of tilt-based input speed. Useful for benchmarking but not implemented in educational systems. The proposed app transforms tilt into one of several usable child-centered interaction methods.
- xvi. **Wobbrock & Myers (2006) – Trackball Text Entry**
Early innovation in motor accessibility through trackball input. However, limited to text and adult usability. This project instead provides multimodal flexibility integrated with learning goals and child safety.
- xvii. **Malu et al. (2018) – Accessible Smartwatch Interactions**
Explores smartwatch interactions for motor impairments. While broadening device form factors, it is not aligned with education. This project situates alternative inputs in a structured educational platform for children.
- xviii. **Chan et al. (2022) – Eye-Tracking Training Improves Learning**
Finds eye-tracking can improve memory and learning outcomes. Strongly relevant but limited to training studies. The proposed app integrates gaze tracking into multimodal educational workflows with motivational and safety layers.
- xix. **Rytterström et al. (2016) – Teachers’ Experiences with Eye-Gaze Computers**
Captures teachers’ experiences of gaze-controlled learning tools. Valuable insights into classroom acceptance but limited to single modality. This project broadens input diversity and introduces customizable, localized design for Sri Lankan primary learners.

xx. Jayasekara et al. (2024) – Smart Wheelchair Robot + Mobile App (Sri_Lanka)

Demonstrates Sri Lankan innovation in accessibility for mobility assistance and home management. While impactful, it does not address educational inclusion. The proposed app fills this gap by applying locally relevant assistive technologies directly to learning environments.

1.3 RESEARCH GAP

Inclusive education emphasizes equal opportunities for all learners, yet children with physical disabilities in primary education (Grades 3–5) continue to face significant challenges when accessing digital learning platforms. Many existing educational applications provide only limited accessibility, with most designed for a single type of disability such as visual or hearing impairments. While these tools are valuable, they often fail to address the complex and diverse needs of children with physical disabilities who may rely on multiple assistive devices and require adaptable, child-friendly interfaces. Furthermore, current solutions rarely incorporate local language support, making it even harder for children in non-English-speaking regions, such as Sri Lanka, to access and benefit from them.

A review of existing literature highlights these gaps clearly. Paulus and Remijn (2021) provide controlled evidence that dwell-time activation is a feasible alternative to manual input, yet this work is confined to performance testing and does not extend to real-world educational systems or safety-enhanced, child-focused design. Similarly, Borgestig and colleagues across multiple studies (2016; 2017 intervention; 2017 parental experiences) demonstrate that gaze-based assistive technology can support children's independence and daily activities. However, these contributions remain focused exclusively on eye-gaze as a single modality, leaving out the integration of complementary controls such as dwell-time customization or voice input, and overlooking the importance of multimodal choice for children with varying motor abilities.

Expanding on these findings, Hsieh et al. (2021) and Hsieh (2024) examine communicative interaction and school-based feasibility of gaze interventions, confirming that gaze can facilitate participation but again limiting exploration to one input pathway. Wilson et al. (2022) validate the acceptability of gaze and switch controls for cognitive testing among children with cerebral palsy, but their work stops short of embedding these methods into a multimodal educational framework. In a related way, Andreassen et al. (2024) and Masayko et al. (2024) capture the psychosocial impact and parental perceptions of gaze-based assistive

technology, offering valuable perspectives on acceptance and quality of life. Still, these studies do not propose mechanisms for integrating voice and dwell alongside gaze to reduce fatigue and increase flexibility for learners. Hemmingsson et al. (2018) illustrate the benefits of early gaze-based intervention for a child with spinal cord injury, but the single-case nature and gaze-only focus again underscore the lack of multimodal strategies.

Other strands of work further highlight fragmentation. Kairy et al. (2021) presents a mobile app to support social participation for people with disabilities, but it is not designed around adaptive learning contexts or multimodal control. Mott et al. (2018) investigate smartphone photography accessibility for motor-impaired users, identifying UI barriers but not addressing learning or child-specific multimodal interaction. Likewise, Cicek et al. (2020) and Fan et al. (2020) propose head-based pointing and eyelid gestures, which innovate on alternative controls but remain isolated technologies without integration into voice or dwell-time frameworks suitable for primary education. Studies on tilt input, such as Castellucci and MacKenzie (2019) and Ljubic et al. (2014), provide insights into no-touch control, but tilt has limited applicability in structured learning for children and is not considered further in this component. Early work like Wobbrock and Myers (2006) on trackball text entry and Malu et al. (2018) on smartwatch interactions for motor-impaired users reinforce the same limitation: creative input methods remain siloed rather than unified in a multimodal, safety-aware, educational platform.

At the same time, research has begun to connect eye-tracking with cognitive and learning benefits. Chan et al. (2022) show that eye-tracking training can improve learning and memory outcomes, confirming the educational potential of gaze input. Yet, when implemented in isolation, gaze risks over-reliance and fatigue, highlighting the need to combine it with voice and dwell-time activation. Teacher perspectives also reflect this gap: Rytterström et al. (2016) document the benefits of gaze-controlled computers in classrooms, but stress that adoption is shaped by ease of use and contextual fit needs that can be better addressed through multimodal, customizable interfaces.

Finally, Sri Lankan work by Jayasekara et al. (2024) demonstrates strong local innovation in accessibility through a smart wheelchair and mobile application. While significant in mobility and home management, it does not address inclusive learning or multimodal interaction for children in primary schools. This highlights a gap between technological development and localized educational inclusion.

Taken together, these studies show that while dwell-time, gaze, and voice inputs have been validated individually, there is still a lack of integrated, child-centered systems. Existing solutions often focus on a single modality, miss safety features,

or fail to support young learners in non-English contexts. This project addresses these gaps by combining dwell-time, gaze tracking, and voice input into a multimodal platform that is safe, engaging, and developmentally appropriate for physically disabled students in Grades 3–5.

Table I. Comparison of Research Gaps in existing studies and Proposed Mobile App

Features Research	Dwell Time	Voice Active	Gaze Track	Multi Option	Child Safety UI	Localization
A Mobile App to Optimize Social Participation [11].						
Longitudinal eye-gaze performance [2].						
Eye-Gaze Control Technology as Early Intervention [9].						
photography accessibility (motor impairments) [12].						
Smart Wheelchair Robot and Mobile Application [21].					Partial	
Our Proposed System						

1.4 RESEARCH PROBLEM

Despite advancements in assistive technology, students with physical disabilities in Grades 3-5 continue to face significant barriers in accessing digital learning tools. Existing solutions suffer from critical limitations:

- **Can fragmented accessibility features be overcome by integrating multiple methods into a single system?**

Most assistive technologies rely on single input methods (e.g., gaze control, switch access, or voice commands) without integrating multi-modal alternatives. This restricts flexibility, forcing users to depend on one interaction mode even when it is inefficient or fatiguing.

- **How can assistive technologies be better adapted to meet the needs of young learners?**

Voice recognition systems are primarily trained on adult speech, leading to inaccuracies for children. Additionally, many assistive tools lack child-friendly interfaces, adjustable settings (e.g., dwell time, button size), and support for local languages.

- **What safety and error-prevention features are necessary to make e-learning tools reliable for physically impaired students?**

Current solutions neglect essential safety features, such as emergency stop commands, accidental input cancellation, or visual feedback mechanisms, increasing frustration and hindering independent use.

- **Is it possible to create a unified educational solution that combines adaptive navigation with curriculum-aligned learning features for young students?**

While some apps address individual disabilities (e.g., switch access for motor impairments or screen readers for visual impairments), none combine adaptive navigation (voice, gaze, facial gestures) with learning-specific needs such as auto-play lessons, simplified UI, and curriculum-aligned content for early education.

This research seeks to bridge these gaps by developing a comprehensive, child-friendly e-learning interface that:

- Integrates multi-modal input (voice, gaze, switches, facial gestures) to reduce dependency on a single control method.
- Prioritizes young learners through age-appropriate voice commands, adjustable interaction settings, and localized language support.

- Ensures safety and usability with emergency stop functions, error cancellation, and visual feedback.
- Supports personalized learning with customizable dwell times, button sizes, and adaptive navigation tailored to individual motor abilities.

2. OBJECTIVES

2.1 MAIN OBJECTIVES

To design and implement an adaptive e-learning mobile interface that enables students with physical disabilities to access and navigate learning content independently, safely, and efficiently.

- **Real Time Data Processing** – Giving the accurate grades for the students' activities when they successfully completed.
- **Providing Collectables (Coins, Trophies, Smile Faces or Stars) for their achievements** – This will encourage students to engage with more Subject related activities.
- **Kids Friendly UI** – This will be easy to use the application without any difficulties especially for the disability students.
- **Providing a friendly AI Assistant** – Using this assistant student will be able to get help when their working at any time.

2.2 SPECIFIC OBJECTIVES

- Integrated multiple control methods, so users can select the comfortable way to use the app.
- Provide customizable settings for dwell time, button size to suit individual needs.
- Implement safety features such as visual countdowns and emergency voice-based stop commands.

3. METHODOLOGY

This research adopts a user-centered, accessibility-focused methodology to design and develop an inclusive, cross-platform e-learning application tailored for students with physical disabilities in grades 3–5. By integrating multiple adaptive input methods, safety features, and customization options, the system aims to bridge the gap between traditional touch-based applications, and the diverse accessibility needs of physically impaired learners. The methodology is divided into six core stages to ensure functionality, usability, and inclusiveness.

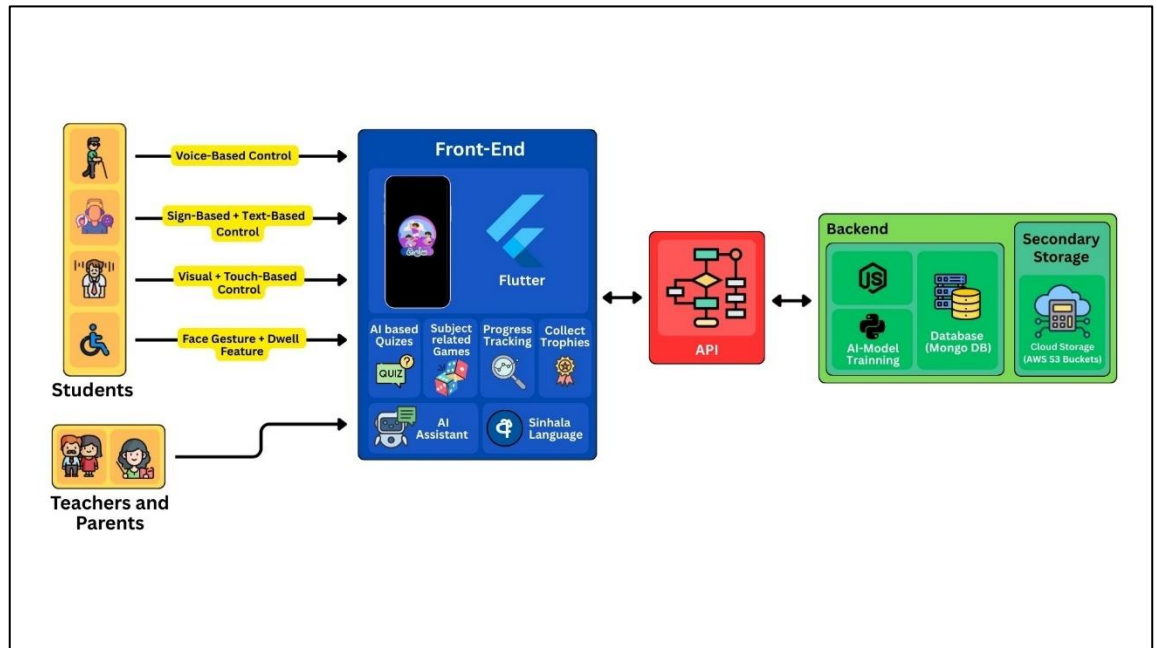


Fig. 1. System Overview Diagram of the complete inclusive mobile learning app

System Design & Architecture

- A modular framework is established to integrate various input methods, including voice control, facial gesture recognition, and dwell-time activation.
- The architecture ensures cross-platform compatibility, functioning seamlessly on Android and iOS devices such as smartphones, tablets, and iPads.
- Accessibility guidelines (WCAG and mobile accessibility standards) are embedded into the design to ensure a universally usable interface.

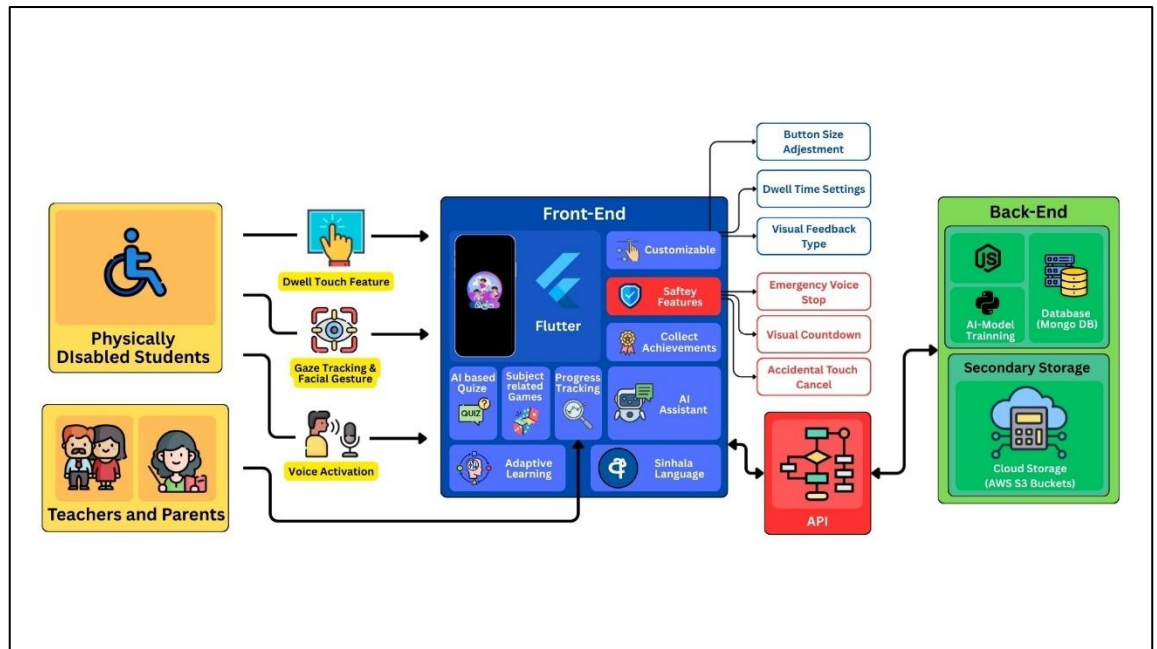


Fig. 2. System Architecture Overview for Physical Impairment Support

4. BUDGET AND BUDGET JUSTIFICATION

Table II. Explains Estimated costs for tools and resources to assist hearing-impaired students.

Table II. Budget Plan for Inclusive mobile learning app for Physical Impairment Support

Purpose	Estimated Cost
UI/UX Redesign Testing	Rs. 5000
Voice Command API	Rs. 10,000
User Testing with Students (Travel & Snacks)	Rs. 5000
AWS services	Rs. 10,000
Total	Rs. 30,000

5. COMMERCIALIZATION ASPECTS OF THE PRODUCT

The proposed solution has clear commercialization potential within the Sri Lankan EdTech market, where over 390,000 students with disabilities face barriers in accessing inclusive education. Currently, most learning applications focus on single disabilities or lack Sinhala language support, creating a strong gap that this product directly addresses.

The application can adopt a freemium model, where core accessibility features (multi-modal inputs, safety functions, Sinhala curriculum alignment) are offered free to ensure inclusivity. Premium features such as advanced analytics for teachers, extended cloud storage, or integration with specialized assistive hardware (eye trackers) can provide revenue opportunities.

Partnerships with schools, NGOs, and government agencies will enhance adoption and scalability, while alignment with the Sri Lankan primary curriculum makes it attractive for direct integration into classrooms. The solution can also scale beyond Grades 3–5 to support higher levels and other disability categories, transforming it into a sustainable educational technology platform.

By addressing a unique, under-served local market with the potential to expand regionally, the system demonstrates strong commercialization and entrepreneurship value.

6. WORK BREAKDOWN STRUCTURE

Table III. Work Breakdown Structure of Physical Impaired Support

Task \ Time	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8
Requirement analysis & Research								
Data gathering, Proposal Doc & Presentation								
Prototype Design								
App Development								
AI related Developments								
Integration & Testing								
Validation & Feedback								
Final Presentation								

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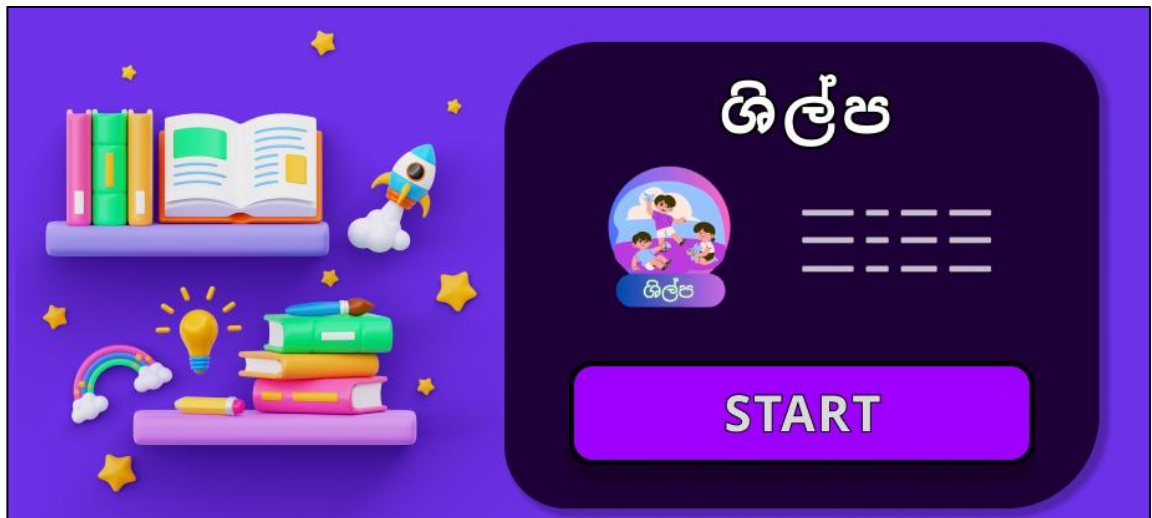
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8. APPENDICES



Appendix A: Application Loading Page Interface



Appendix B: Select Disability Type Page



Quiz

පාඩම: සරල බෙදීම

කාලය: 15min



SUBMIT

Appendix C: AI generated Quiz Page



බෙදීම

පාඩම: සරල බෙදීම

කාලය: 15min

Enroll

Quiz

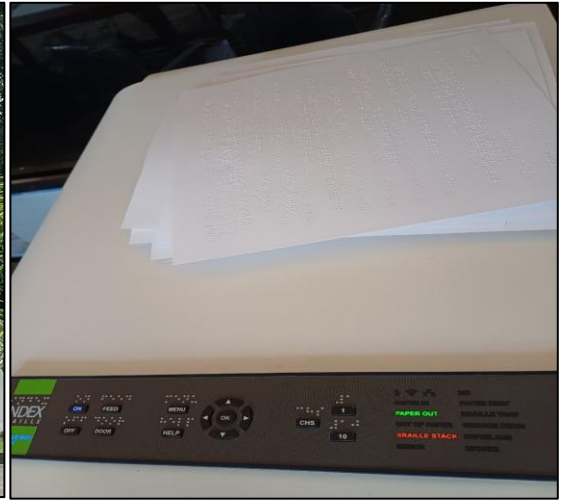
Related Games







Appendix D: Content Page



Appendix E: Field visit at NIE (National Institute of Education)